

- 9 The engine of a car has maximum output power of 54 kW. The air resistance acting on the car when it is moving with speed v is cv^2 , where c is 2.0 kg m^{-1} . What is the maximum speed that can be achieved by the car on level road?

A 3.0 m s^{-1}

B 5.0 m s^{-1}



C 30 m s^{-1}

D 160 m s^{-1}

At max speed, rate of work done against air resistance = $54 \times 10^3 \text{ W}$

$$Fv = cv^2 \times v = 54 \times 10^3 \rightarrow v = 30 \text{ m s}^{-1}$$

10 A pendulum bob at the end of the string which is fixed at O moves in a horizontal circle on the